

Study BM or GAUSSIAN
PROCESS BEFORE!

— Existence of Brownian Motion —

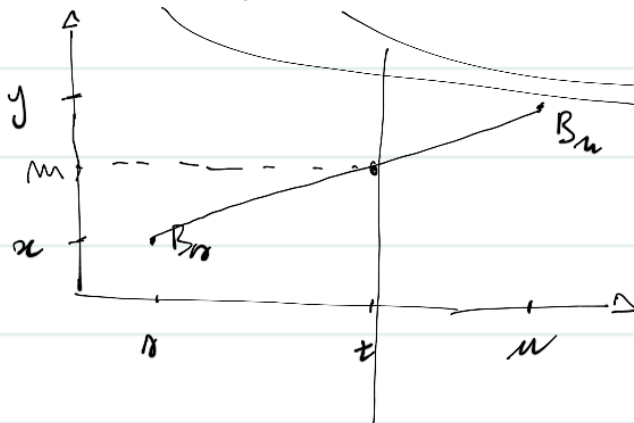
Lévy's construction.

There is a useful result we should prove in advance,
which helps the intuition. PROP 3.1

Proposition. Let $(B_t)_{t \geq 0}$ be a 1-dim BM and $0 \leq r < t < u$.
Then

$$A \mapsto P(B_t \in A \mid B_r = x, B_u = y) = N(m, \sigma^2)$$

$$m := \frac{u-t}{u-r}x + \frac{t-r}{u-r}y \quad \sigma^2 := \frac{(u-t)(t-r)}{u-r}$$



"weights" for x and y :
if $u-t = t-r$ then
 $m = \frac{x+y}{2}$

m is on the line !!

$$B_t \sim N(m, \sigma^2)$$

Proof. Use that the vector (B_r, B_t, B_u) is Gaussian
with mean $\underline{0}$ and covariance matrix C given by

$$C := \begin{bmatrix} r & r & r \\ r & t & t \\ r & t & u \end{bmatrix}$$

(Prove this as an exercise).

Use that if (x, y, z) is a random vector with density $p_{x,y,z}(x, y, z)$ and $p_{y,z}(y, z)$ is the marginal of (y, z) , then

$$P(X \in B | Y=y, Z=z) = \int \mathbb{1}_B(x) p_{x|y,z}(x) dx$$

where

$$p_{x|y,z}(x) = \frac{p_{x,y,z}(x, y, z)}{p_{y,z}(y, z)}.$$

In our case the distribution of (B_r, B_t, B_u) is

$$p_{r,t,u}(x, y, z) = \frac{1}{(2\pi)^{3/2}} \frac{1}{\sqrt{(u-t)(t-r)r}} \exp\left(-\frac{1}{2} \left[\frac{(z-y)^2}{u-t} + \frac{(y-x)^2}{t-r} + \frac{z^2}{r} \right]\right)$$

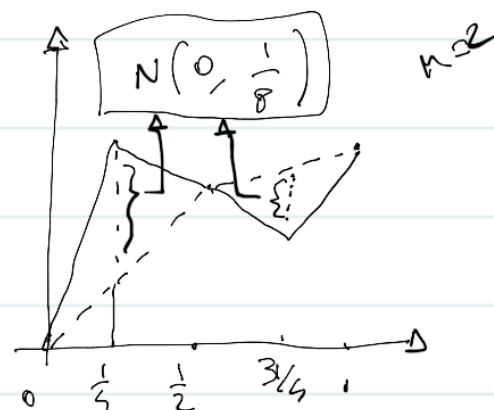
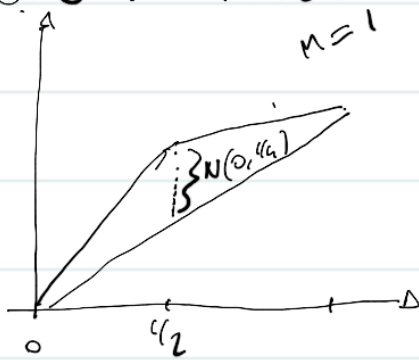
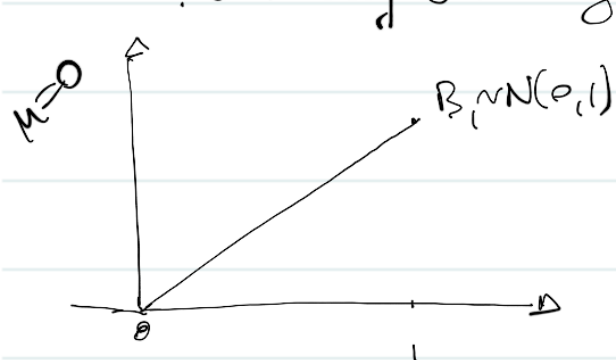
and

$$p_{r,u}(x, z) = \frac{1}{(2\pi)} \frac{1}{\sqrt{(u-r)r}} \exp\left(-\frac{1}{2} \left[\frac{(z-x)^2}{u-r} + \frac{z^2}{r} \right]\right).$$

Do this as an exercise.



The idea of the Lévy construction is to define BM iteratively on dyadic numbers



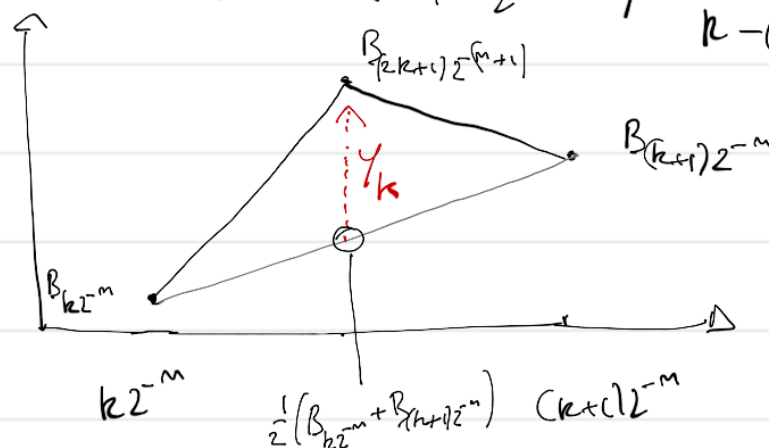
By the previous proposition we have that
 $A \mapsto P(B_{(2k+1)2^{-m}} \in A \mid B_{k2^{-m}} = x, B_{(k+1)2^{-m}} = y) = N\left(\frac{1}{2}(x+y), \frac{1}{4}2^{-m}\right)$

Here is the Algorithm:

1°. Set $B_0 = 0$ and $B_1 \sim N(0, 1)$

2°. Assume we have already defined $B(k2^{-m}), k=0 \dots 2^m$, such that
 $B_{(k+1)2^{-m}} - B_{k2^{-m}} \sim N(0, 2^{-m})$ and are independent ($k=0 \dots 2^m-1$)

3°. Denote γ_k the distance between the interpolation of
 $B_{(k+1)2^{-m}}$ and $B_{k2^{-m}}$ at the point $(2k+1)2^{-(m+1)}$
 and suppose $\gamma_k \sim N\left(0, \frac{1}{2^{m+2}}\right)$ and are independent
 $k=(0 \dots 2^m-1)$



$$\gamma_k \sim N\left(0, \frac{1}{2^{m+2}}\right)$$

$$k=0 \dots 2^m-1$$

and define now

$$B_{e2^{-(m+1)}} := \begin{cases} B_{k2^{-m}} & e=2k \text{ (even)} \\ \frac{1}{2}(B_{k2^{-m}} + B_{(k+1)2^{-m}}) + \gamma_k & e=2k+1 \text{ (odd)} \end{cases}$$

$e=2k$ (even)

$$\boxed{e=2k+1 \text{ (odd)}}$$

$\hookrightarrow$ new dyadic when m increases

RMK 3.3

Remark. Take the increments $B_{(2k+1)2^{-(m+1)}} - B_{2k2^{-(m+1)}} =$
 $= \frac{1}{2}(B_{k2^{-m}} + B_{(k+1)2^{-m}}) + \gamma_k - B_{k2^{-m}} = \frac{1}{2}(B_{(k+1)2^{-m}} - B_{k2^{-m}}) + \gamma_k$
 and note that they are again Gaussian and indep. by
 $\text{Var}(\gamma_k) = \frac{1}{4} \cdot \frac{1}{2^m} + \frac{1}{2^{m+2}} = \frac{1}{2^{m+2}}$

properties of Gaussian r.v.'s:

- $\exists (X_k)_{k=1 \dots N} \quad (Y_k)_{k=1 \dots N}$ such that
- $X_k \sim N(0, \sigma^2) \quad Y_k \sim N(0, \sigma^2)$
 - $X_1 \dots X_N$ are indep $Y_1 \dots Y_N$ are indep.
 - $(X_k)_{k=1 \dots N}$ and $(Y_k)_{k=1 \dots N}$ are indep.

Then $X_1 + Y_1, X_1 - Y_1, \dots, X_N + Y_N, X_N - Y_N$ are indep and
 $X_k \pm Y_k \sim N(0, 2\sigma^2) \quad k=1 \dots N.$

To apply this in our situation take $X_k = \frac{1}{2}(B_{(k+1)2^{-n}} - B_{k2^{-n}})$
 $Y_k = Y_k$

THM 3.4

Here is the result by Lévy.

Theorem (Lévy 1940) For $n \geq 0$, let $W_n(t, \omega)$ be the linear interpolation of $B_{k2^{-n}} \quad k=0 \dots 2^n$. Then the series

$$\bar{B}(t, \omega) := \sum_{n=0}^{\infty} (W_{n+1}(t, \omega) - W_n(t, \omega)) + W_0(t, \omega) \quad t \in [0, 1]$$

converges a.s. uniformly in $t \in [0, 1]$. The process $(\bar{B}_t)_{t \in [0, 1]}$ is (indistinguishable from) a BM

Note that

$$\begin{aligned} \bar{B}(t, \omega) &:= \sum_{n=0}^{\infty} (W_{n+1}(t, \omega) - W_n(t, \omega)) + W_0(t, \omega) \\ &= \lim_{N \rightarrow \infty} \sum_{n=0}^N (W_{n+1}(t, \omega) - W_n(t, \omega)) + W_0(t, \omega) \\ &= \lim_{N \rightarrow \infty} W_N(t, \omega) \end{aligned}$$

Definition. Let $(X_t)_{t \geq 0}, (Y_t)_{t \geq 0}$ be real valued random processes on a given $(\Omega, \mathcal{F}, \mathbb{P})$.

- 1) The processes are called indistinguishable if there is a null set $\Omega_0 \subset \Omega$ s.t.

$$X_t(\omega) = Y_t(\omega) \quad \forall \omega \notin \Omega_0, \forall t \geq 0$$

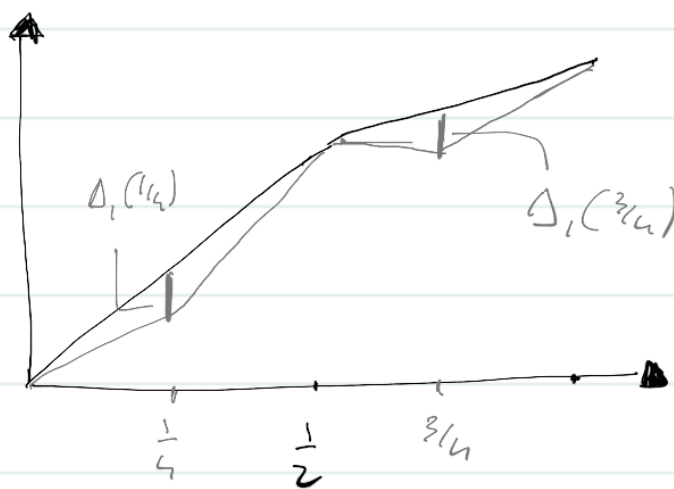
- 2) The processes are called versions of each other if $\forall t \geq 0$ there exist a null set $\Omega_0(t) \subset \Omega$ s.t.

$$X_t(\omega) = Y_t(\omega) \quad \forall \omega \notin \Omega_0(t)$$

Proof. Set $\Delta_n(t, \omega) := W_{n+1}(t, \omega) - W_n(t, \omega)$.

By construction, the maximal distance is obtained at one of the midpoints of the intervals $[k2^{-n}, (k+1)2^{-n}]$:

$$\sup_{t \in [0, 1]} |\Delta_n(t, \omega)| \leq \sup_{1 \leq k \leq 2^n} |\Delta_n((2k-1)2^{-n-1}, \omega)|$$



$$n=2$$

We show now that

$$\sum_{m=1}^{\infty} \sup_{1 \leq k \leq 2^m} |\Delta_m((2k-1)2^{-m-1}, \omega)| < \infty$$

on a set $\Omega_0 \subset \Omega$ with $P(\Omega_0) = 1$. This implies (a.p.) uniform convergence of the series.

Let y_m be such that $\sum_m y_m < \infty$.

Note that

$$\sum_m P\left(\sup_{1 \leq k \leq 2^m} |\Delta_m((2k-1)2^{-m-1})| > y_m\right) \\ \sim N(0, 2^{-(m+2)}) \longrightarrow \text{these are the } y_m$$

$$\leq \sum_m P\left(\left|\frac{1}{\sqrt{2^{m+2}}} \Delta_m((2k-1)2^{-m-1})\right| > \sqrt{2^{m+2}} y_m\right)$$

$$P(\sup_{k=1}^{2^m} U_k > y) \leq \sum_{k=1}^{2^m} P(U_k > y) \\ \leq 2^m \int_{\sqrt{2^{m+2}} y_m}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy$$

$$\leq 2^{m+1} \int_{\sqrt{2^{m+2}} y_m}^{\infty} \frac{y}{\sqrt{2\pi}} e^{-y^2/2} dy > 1$$

$$= 2^{m+1} \frac{1}{\sqrt{2^{m+2}} y_m} \frac{1}{\sqrt{2\pi}} e^{-\frac{y_m^2 2^{m+2}}{2}}$$

choose $y_m := \frac{c \sqrt{2^m \log 2}}{\sqrt{2^{m+2}}}$ for some $c > 1$

$$= \frac{2^{m+1}}{\sqrt{2\pi}} \frac{1}{c \sqrt{2m \log 2}} e^{-c^2 2^m \log 2}$$

$$= \frac{1}{\sqrt{2\pi}} \frac{1}{c} \frac{1}{\sqrt{m}} \frac{1}{\sqrt{\log 2}} \frac{2^m}{2^m c^2} \leq \frac{1}{\sqrt{m \log 2}} \frac{1}{2^{m(2c^2-1)}}$$

ad B

$$\sum_n \mathbb{P} \left(\underbrace{\max_{1 \leq k \leq 2^m} \left| \Delta_n((z_{k-1}) z^{-m-1}) \right|}_{=: A_n} > y_n \right) < \infty.$$

By the 1st B-C lemma $\mathbb{P}(\limsup A_n) = 0 \rightarrow \mathbb{P}(\liminf A_n^c) = 1$,
hence there exist $\Omega_0 \subset \Omega$ with $\mathbb{P}(\Omega_0) = 1$ a.s. $\stackrel{=}{=} \Omega_0$

$\forall \omega \in \Omega_0 \exists N(\omega) \in \mathbb{N} : \forall n > N(\omega)$ it is true that

$$\max_{1 \leq k \leq 2^m} \left| \Delta_n((z_{k-1}) z^{-m-1}) \right| \leq y_n = \frac{c \sqrt{2m \log 2}}{\sqrt{2^{m+2}}}$$

It follows that $\lim_{n \rightarrow \infty} W_n(t, \omega)$ exists for all $\omega \in \Omega_0$
 $t \in [0, 1]$, and the series converges uniformly in $t \in [0, 1]$.

In particular, the paths $t \mapsto B(t, \omega)$ are continuous
for all $\omega \in \Omega_0$ by uniform continuity.

Now we prove that

$$\tilde{B}(t, \omega) := \begin{cases} \bar{B}(t, \omega), & \omega \in \Omega_0, \\ 0, & \omega \notin \Omega_0, \end{cases} \quad t \in [0, 1],$$

is a Brownian motion.

It is clear that $\tilde{B}(0) = 0$ and that $(\tilde{B}_t)_{t \in [0, 1]}$

her continuous sample paths.

Stationarity:

Let $0 \leq s \leq t$ be two dyadic numbers, i.e., $s = j2^{-n}$ $t = k2^{-n}$ for some $j < k$ and $n \in \mathbb{N}$.

$$\tilde{B}(k2^{-n}) - \tilde{B}(j2^{-n}) \stackrel{a.s.}{=} \bar{B}(k2^{-n}) - \bar{B}(j2^{-n})$$

$$= W_n(k2^{-n}) - W_n(j2^{-n})$$

$$= \sum_{\ell=j+1}^k \left(B(\ell 2^{-n}) - B((\ell-1) 2^{-n}) \right)$$

indep for $\ell = j+1, \dots, k$

$\sim N(0, t-s)$

$\begin{matrix} \nearrow \\ \frac{k-j}{2^n} 2^{-n} = \frac{k-j}{2^n} \end{matrix}$
 $\begin{matrix} B(\ell 2^{-n}) - B((\ell-1) 2^{-n}) \\ \sim N(0, 2^{-n}) \end{matrix}$

For general $s, t \in [0, 1]$ take two sequences of dyadic numbers s_k, t_k with $t_k \downarrow t$, $s_k \downarrow s$. Then

$$\begin{aligned} \mathbb{E} e^{i\lambda (\tilde{B}(t) - \tilde{B}(s))} &= \lim_{k \rightarrow \infty} \mathbb{E} \exp(i\lambda (\tilde{B}(t_k) - \tilde{B}(s_k))) \longrightarrow \text{Bounded Convergence} \\ &= \lim_{k \rightarrow \infty} e^{-\frac{t_k - s_k}{2} \lambda^2} = e^{-\frac{t-s}{2} \lambda^2} \end{aligned}$$

Independence follows analogously, using indep. of increments on the dyadic numbers.

$$\mathbb{E} e^{i\lambda [\tilde{B}(t_1) - \tilde{B}(t_0)] [\tilde{B}(s_1) - \tilde{B}(s_0)]} = \lim_k$$

$$= \dots = \mathbb{E} e^{i\lambda [\tilde{B}(t_1) - \tilde{B}(t_0)]} \mathbb{E} e^{i\lambda [\tilde{B}(s_1) - \tilde{B}(s_0)]}$$

Now we use this result to construct a 1-dim BM on $t \geq 0$.
 Gsibuy. There is a 1-dim BM $(B_t)_{t \geq 0}$.

Proof. Let $(\Omega_k, \mathcal{A}_k, P_k)$, $k \in \mathbb{N}_0$, be probability spaces. Use the previous theorem to construct ~~BM~~ BMs $(B_t^k)_{t \in [0,1]}$ on $(\Omega_k, \mathcal{A}_k, P_k)$. Consider the product space $(\Omega, \mathcal{A}, P) := \bigotimes_{k \in \mathbb{N}_0} (\Omega_k, \mathcal{A}_k, P_k)$.

Define, on Ω , the stochastic processes

$$W^k(t, \omega) := B^k(t, \omega_k), \quad t \in [0,1], k \in \mathbb{N}_0$$

where $\Omega \ni \omega = (\omega_0, \omega_1, \dots, \omega_k, \dots)$

The processes $(W^k(t))_{t \in [0,1]}$, $k \in \mathbb{N}_0$, are independent BMs on Ω .

Now define

$$B_t := \begin{cases} W^0(t) & t \in [0,1] \\ W^0(1) + W^1(t-1) & t \in [1,2] \\ \sum_{k=0}^{m-1} W^k(1) + W^m(t-m) & t \in [m, m+1], m \in \mathbb{N}_0 \end{cases}$$

We prove that $(B_t)_{t \geq 0}$ is a BM on $(\Omega, \mathcal{A}, P)$.

Continuity of paths is clear.

Stationarity of increments: let $s < t$ be such that $s \in [l, l+1)$ and $t \in [m, m+1)$ for some $l \leq m$.

If $l = m$, by definition

$$B_t - B_s = W^m(t-m) - W^m(s-m) \sim N(0, t-s)$$

If $e \leq m$, we have

$$\begin{aligned}
 B_t - B_r &= \left(\sum_{k=0}^{m-1} W^k(1) + W^m(t-m) - \left(\sum_{k=0}^{e-1} W^k(1) + W^e(r-e) \right) \right) \\
 &= \sum_{k=e}^{m-1} W^k(1) + W^m(t-m) - W^e(r-e) \\
 &= \underbrace{W^e(1) - W^e(r-e)}_{\sim N(0, 1-(r-e)) \text{ since } W^e \text{ is BM}} + \underbrace{\sum_{k=e+1}^{m-1} W^k(1) + W^m(t-m)}_{\sim N(0, m-e-1) \text{ since } W^k \text{ are indep } N(0,1)} \\
 &\quad \sim N\left(1-(r-e) + m-e-1 + t-m, t-r\right)
 \end{aligned}$$

We check independence of increments (all's do this only for two increments). Take r & t on with $r \in [e, e+1)$, $t \in [m, m+1)$ and $u \in [n, n+1)$. Then

$$B_u - B_t = \begin{cases} W^m(u-m) - W^m(t-m) & m \geq n \\ W^m(1) - W^m(t-m) + \sum_{k=m+1}^{n-1} W^k(1) + W^n(u-n) & m < n \end{cases}$$

We wrote before the increment $B_t - B_r$. The r.v. appearing in the representations for $B_u - B_t$ and $B_t - B_r$ are indep. and thus $B_t - B_r$ and $B_u - B_t$ are independent.

The Canonical Brownian Motion

So far we constructed a process $(B_t)_{t \geq 0}$ which is a BM on a given probability space. The canonical construction of BM, instead, aims to construct a probability space such that a given process (the canonical process) is a BM.

DEF 3.7

Let $I := [0, \infty)$ and

$$C_{0,1}(I) := \{w: I \rightarrow \mathbb{R} : w \text{ is continuous and } w(0) = 0\}.$$

The canonical process, $(X_t)_{t \geq 0}$, is defined by

$$X_t(w) := w(t), \quad w \in C_0(I).$$

So our goal is to find a probability measure μ on (a suitable σ -Algebra on) $\Omega = C_{0,1}(I)$ s.t. $(X_t)_{t \geq 0}$ is a BM. Which σ -Algebra do we choose?

→ Consider $C_{0,1}(I)$ as a metric space with the metric $\rho(r, w) := \sum_{n=1}^{\infty} 2^{-n} \min \left\{ n \rho \left[\sup_{0 \leq t \leq n} |r(t) - w(t)|, 1 \right], 1 \right\}.$

One can prove that, by definition,

$$\rho(r_n, r) \rightarrow 0 \iff r_n \rightarrow r \text{ uniformly on compact sets} \\ \text{(i.e., } \forall T > 0, \lim_{n \rightarrow \infty} \sup_{t \in [0, T]} |r_n(t) - r(t)| = 0)$$

It is also true that the metric space is separable, i.e., there exists a countable dense subset. The Borel σ -Algebra on this space is the smallest σ -Algebra containing the open sets $U \subset C_0$, i.e., the sets U s.t. for every $u \in U$ there exists $r > 0$ s.t.

$$B_p(u, r) := \{v \in C_0(\mathbb{I}) ; p(u, v) < r\} \subset U.$$

open ball in the metric space $(C_0(\mathbb{I}), p)$

→ Another possibility is to take, on the space

$$\mathcal{R}^{\mathbb{I}} = \{w: \mathbb{I} \rightarrow \mathcal{R}\}$$

the product σ -Algebra, i.e.,

$$\mathcal{B}^{\mathbb{I}}(\mathcal{R}) := \sigma(\pi_t^{-1}(B) : B \in \mathcal{B}(\mathcal{R}), t \geq 0) \\ = \sigma(\pi_t : t \geq 0)$$

$$\mathcal{B}^{\mathbb{I}}(\mathcal{R}) := \sigma(\text{measurable rectangles})$$

$\{w \in \mathcal{R}^{\mathbb{I}} : w(t) \in A_i, A_i \neq \emptyset \text{ for a finite number of } t, A_i \in \mathcal{B}(\mathcal{R})\}$

generated by the projections $\pi_t: \mathcal{R}^{\mathbb{I}} \rightarrow \mathcal{R} \quad w \mapsto w(t)$.

Since $C_0(\mathbb{I}) \subset \mathcal{R}^{\mathbb{I}}$ the trace of $\mathcal{B}^{\mathbb{I}}(\mathcal{R})$ on $C_0(\mathbb{I})$ is

$$C_0(\mathbb{I}) \cap \mathcal{B}^{\mathbb{I}}(\mathcal{R}) := \{B \cap C_0(\mathbb{I}) ; B \in \mathcal{B}^{\mathbb{I}}(\mathcal{R})\}$$

Recall: the trace is a σ -Algebra.

What is the best choice? LEMMA 3.8

Lemma. $\mathcal{B}(C_0(\mathbb{I})) = C_0(\mathbb{I}) \cap \mathcal{B}^{\mathbb{I}}(\mathcal{R})$.

Proof. The mapping $C_0 \ni w \mapsto \pi_t(w) = w(t)$

is continuous wrt ρ . Indeed if $t \in [0, m]$ for some m , then

$$\begin{aligned} \min \{1, |\pi_t(w) - \pi_t(v)|\} &= \min \{1, |w(t) - v(t)|\} \\ &\leq \min \left\{1, \min_{n \leq m} |w(n) - v(n)|\right\} \\ &= 2^m 2^{-m} \min \left\{1, \min_{n \leq m} |w(n) - v(n)|\right\} \\ &\geq 2^m \sum_k 2^{-k} \min \left\{1, \min_{n \leq k} |w(n) - v(n)|\right\} \\ &= 2^m \rho(v, w) \end{aligned}$$

It follows that, $\forall \varepsilon \exists \delta_\varepsilon : \rho(v, w) < \delta_\varepsilon \implies |\pi_t(w) - \pi_t(v)| < \varepsilon$. $\implies \delta_\varepsilon = \frac{\varepsilon}{2^m}$

Therefore $\pi_t: (C_0(I), \mathcal{B}(C_0(I))) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable since any continuous function between (Borel) measure space is measurable. It follows that

$$C_0(I) \cap \mathcal{B}^F(\mathbb{R}) = \sigma(\pi_t|_{C_0(I)} : t \geq 0) \subset \mathcal{B}(C_0(I)).$$

Now we prove the opposite inclusion.

Since $(C_0(I), \rho)$ is a separable metric space then the Borel σ -Algebra is generated by the open balls $B_\rho(w, r)$, $w \in C_0(I)$, $r > 0$. Therefore it is sufficient to show that $B_\rho(w, r) \in C_0(I) \cap \mathcal{B}^F(\mathbb{R})$ for all $w \in C_0$, and $r > 0$. We have

$$\begin{aligned} \rho(v, w) &= \sum_{n \geq 1} 2^{-n} \min \left\{1, \min_{0 \leq t \leq n} |\pi_t(w) - \pi_t(v)|\right\} \\ &= \sum_{n \geq 1} 2^{-n} \min \left\{1, \min_{t \in [0, n] \cap \mathbb{Q}} |\pi_t(w) - \pi_t(v)|\right\} \end{aligned}$$

so that

$$C_0(\mathbb{I}) \ni v \mapsto F(v) := p(v, w) \in \mathbb{R} \text{ is measurable}$$

w.r.t. $\sigma(\mathcal{W}_t | C_0(\mathbb{I}) : t \geq 0) = \mathcal{B}^{\mathbb{I}}(\mathbb{R}) \cap C_0(\mathbb{I})$.

(by properties of measurable functions)
 $\uparrow$
 measurable
 $\downarrow$
 RECALL
 $\mathcal{W}_t$ is a MEAS!

Therefore

$$B_p(w, r) = F^{-1}(C_0(r)) \in \mathcal{B}^{\mathbb{I}}(\mathbb{R}) \cap C_0(\mathbb{I})$$

We remark that, by definition, the sets

$$\{\mathcal{W}_{t_1} \in A_1, \dots, \mathcal{W}_{t_m} \in A_m\}, \quad t_i \in \mathbb{I}, A_i \in \mathcal{B}(\mathbb{R}), m \in \mathbb{N}$$

are a generator of $\mathcal{B}^{\mathbb{I}}(\mathbb{R})$. This implies the following:

Corollary. Every probability measure on $(C_0(\mathbb{I}), \mathcal{B}(C_0(\mathbb{I})))$ is uniquely determined by its values on the cylinder sets

COR 3.9

$$\{w \in C_0(\mathbb{I}) : w(t_1) \in A_1, \dots, w(t_m) \in A_m\}, \quad t_i \in \mathbb{I}, A_i \in \mathcal{B}(\mathbb{R}), m \in \mathbb{N}$$

Proof. Recall that $\mathcal{W}_t: \mathbb{R}^{\mathbb{I}} \rightarrow \mathbb{R}, w \mapsto w(t)$.

By definition the sets

$$\{\mathcal{W}_{t_1} \in A_1, \dots, \mathcal{W}_{t_m} \in A_m\}, \quad t_i \in \mathbb{I}, A_i \in \mathcal{B}(\mathbb{R}), m \in \mathbb{N},$$

are a generator of $\mathcal{B}^{\mathbb{I}}(\mathbb{R})$. Therefore

$$C_0(\mathbb{I}) \cap \{\mathcal{W}_{t_1} \in A_1, \dots, \mathcal{W}_{t_m} \in A_m\} = \{w \in C_0(\mathbb{I}) : \mathcal{W}_{t_1} \in A_1, \dots, \mathcal{W}_{t_m} \in A_m\}$$

$t_i \in \mathbb{I}$
 $A_i \in \mathcal{B}(\mathbb{R})$
 $m \in \mathbb{N}$

generator $C_0 \cap \mathcal{B}^{\mathbb{I}}(\mathbb{R}) \stackrel{\uparrow}{=} \mathcal{B}(C_0(\mathbb{I}))$
 by previous result

The generator above is a $\mathbb{R}$ -system and therefore if two probabs, measures coincide on it, they coincide on $(C_{(0)}(\mathbb{I}), B(C_{(0)}(\mathbb{I})))$.

RMK 3.10

Remark. The σ -Algebra $B^{\mathbb{I}}(\mathbb{R})$ is "small".
For example it is possible to prove that

- For every $I \in B^{\mathbb{I}}(\mathbb{R})$, there exists an at most countable set $S \subset I$ s.t. $\left\{ \begin{array}{l} \text{of increments in } B^{\mathbb{I}} \\ \text{Coincidence also in inverse measurable} \end{array} \right\}$
 $v \in I, w \in \mathbb{R}^{\mathbb{I}}, v|_S = w|_S \implies w \in I$. $\textcircled{A}$

- $C_{(0)}(\mathbb{I}) \notin B^{\mathbb{I}}(\mathbb{R})$
 $\hookrightarrow$ here τ continuous along $\exists w: w|_S = \tau|_S$ even non Gaussian process

Sketch of proof. Show that $\Sigma' := \{ I \in B^{\mathbb{I}}(\mathbb{R}) : I \text{ satisfies } \textcircled{A} \text{ for an at most countable set } S \subset I \}$

$\downarrow$ it means that the sets in $B^{\mathbb{I}}(\mathbb{R})$ are big!

is a σ -Algebra. Check that the family of cylinder sets is in Σ' and conclude that

$$\sigma(\mathcal{G}) (= B^{\mathbb{I}}(\mathbb{R})) \subset \Sigma'$$

Use the first assertion to prove the second.

Now we go back to the construction of a probability space so that the canonical process is a BM. For this, we resort now to the Kolmogorov's Theorem.

Theorem (Kolmogorov). For each $t_1, \dots, t_n \geq 0, n \in \mathbb{N}$, with $t_i \neq t_j$ for $i \neq j$ let $p_{t_1, \dots, t_n}$ be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Assume that the family of probability measures is consistent, i.e.,

- 1) $p_{t_1, \dots, t_n}(A_1 \times \dots \times A_n) = p_{t_{\sigma(1)}, \dots, t_{\sigma(n)}}(A_{\sigma(1)} \times \dots \times A_{\sigma(n)})$
for every permutation $\sigma: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ and $A_i \in \mathcal{B}(\mathbb{R})$
- 2) $p_{t_1, \dots, t_n}(A_1 \times \dots \times A_{n-1} \times \mathbb{R}) = p_{t_1, \dots, t_{n-1}}(A_1 \times \dots \times A_{n-1})$
for all $A_i \in \mathcal{B}(\mathbb{R})$.

Then there exists a unique probability measure μ on $(\mathbb{R}^{\mathbb{R}}, \mathcal{B}^{\mathbb{R}}(\mathbb{R}))$ such that the canonical process $X_t(\omega) = \omega(t)$ has finite dimensional distribution $p_{t_1, \dots, t_n}$ i.e.

$$P(X_{t_1} \in A_1, \dots, X_{t_n} \in A_n) = p_{t_1, \dots, t_n}(A_1 \times \dots \times A_n)$$

for all $t_i \geq 0, A_i \in \mathcal{B}(\mathbb{R})$.

Proof. See References

COR 3.12

Corollary. There exists a unique measure μ on $(\mathbb{R}^{\mathbb{R}}, \mathcal{B}^{\mathbb{R}}(\mathbb{R}))$ such that the canonical process $(X_t)_{t \geq 0}$ satisfies: $X_0 = 0$, stationary increments, indep. increments.

Proof. For $t_1, \dots, t_n \geq 0$, with $t_i \neq t_j, i \neq j$, let $p_{t_1, \dots, t_n}$ be the Gaussian distribution with mean vector $\underline{m} = \underline{0}$ and covariance matrix $C: (C)_{ij} = \min(t_i, t_j), i, j = 1, \dots, n$. This probability measure is consistent. Therefore there

exists a unique probability measure μ on $(\mathbb{Q}^{\mathbb{R}}, \mathcal{G}^{\mathbb{R}}(\mathbb{R}))$ such that the canonical process $(X_t)_{t \geq 0}$, satisfies

$$(X_{t_1}, \dots, X_{t_n}) \stackrel{\Delta}{=} P_{t_1, \dots, t_n}(\cdot) = N(\underline{0}, C)$$

for all $t_i \geq 0$. The result follows by the following result

LEMMA 3.13

Lemma. Let $(X_t)_{t \geq 0}$ be a 1-dim stochastic process. The following are equivalent

- (i) $X_0 = 0$, it has stationary and indep increments
- (ii) The distribution of the vector $(X_{t_1}, \dots, X_{t_n})$, is, for any choice of $t_1, \dots, t_n \geq 0$, Gaussian with mean $\underline{0}$ and covariance matrix $n \times n$ C .

In particular

$$E e^{i \langle \xi, X \rangle} = e^{-\frac{1}{2} \langle \xi, C \xi \rangle} \quad \xi \in \mathbb{R}^n$$

What is missing? Continuity of sample paths!

THM 3.14

We resort to the following theorem (Kolmogorov - Centsov)

Let $(X_t)_{t \geq 0}$ be a d -dimensional stochastic process on a probability space $(\Omega, \mathcal{A}, P)$. If there are constants $c > 0$ and $\alpha, \beta > 0$ st.

$$E(|X_t - X_s|^\alpha) \leq \frac{c}{\alpha} |t - s|^{1+\beta}, \quad P, t \geq 0,$$

then X_t has a version with continuous sample paths. In particular the version can be chosen so that the

paths are almost surely locally Hölder continuous of order δ , for $\delta \in (0, \frac{1}{2})$, i.e., for a.e. $\omega \in \Omega$ there is, for every $T > 0$ and $\delta \in (0, \frac{1}{2})$ some constant $c = c(\delta, T)$ such that

$$|Y_t^{(\omega)} - Y_s^{(\omega)}| \leq c |t - s|^\delta, \quad s, t \in [0, T]$$

Proof. See reference

COR 3.15

Corollary. Brownian motion exists.

Proof. We know that the canonical process on $(\mathbb{R}^d, \mathcal{B}^d(\mathbb{R}), \mu)$ satisfies $X_0 = 0$ and has stationary and indep increments. Then, note that

$$\begin{aligned} \mathbb{E} |X_t - X_s|^4 &= \mathbb{E} |X_{t-s}|^4 \stackrel{\text{stationary}}{=} |t-s|^2 \mathbb{E} |X_1|^4 < \infty \\ &\quad \downarrow \text{self-similarity} \\ &\quad X_{t-s} \sim \sqrt{t-s} X_1 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} d=3$$

for all $s, t \geq 0$. Hence the version (say $B_{t,t \geq 0}$) given by Kolmogorov-Centsov provides a process with continuous paths.

RMK 3.16

Remark. Let $B = (B_t, t \geq 0)$ be the continuous version. Then

$$(\mathbb{R}^d, \mathcal{B}^d(\mathbb{R}), \mu) \xrightarrow{B} (C_{[0,1]}(\mathbb{R}^d), \mathcal{B}(C_{[0,1]}(\mathbb{R}^d)), P)$$

where $P(B) = \mu(\bar{A})$ for any $\bar{A} \in \mathcal{B}^d: \bar{A} \cap C_0 = B$. The canonical

process on $(C([0,1]), \mathcal{B}(C([0,1])), P)$ is the BM we were looking for.

The d -dimensional BM

It is possible to construct a process fitting the definition of BM in $\mathbb{R}^d$. This is called the d -dimensional BM.

Essentially, the procedure is based on the following implications

$B_t := (B_t^{(1)} \dots B_t^{(d)})$ is a d -dimensional Brownian Motion



The coordinate processes $(B_t^{(j)})_{t \geq 0}, j=1 \dots d$ are indep. 1dim. BMs.

The implications above can be proved, e.g., using characteristic functions.